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WHITING SCHOOL
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FPGA Design Using VHDL

Module 4A: Hardware vs. Simulation

This Module, You Should...

- Compare when VHDL simulation does not match hardware
- Explain how a **simulator evaluates statements in a process**
- Recall what are VHDL **Objects** versus **Types**
- Distinguish between the different **Types**
- Declare **user-defined** types

Simulate vs. Hardware



- If design works in simulation, will it work in hardware?
 - No, not necessarily
- Example – One PPS counter
- Looks good in simulation
- But what happens when synthesized and mapped to FPGA

Behavioral Description

- Process that generates a pulse every one second.
 - Create a 26-bit **counter** that increments by one every **rising** edge of a 50 MHz **clock** and clears to zero when the maximum count reaches **50×10^6** then continues counting again. The counter is triggered by the **clock** and **asynchronously** reset by the "reset" signal.
 - Create a CSA for a signal that pulses **high** when counter reaches **50×10^6** , else will remain **low**. The frequency of the pulse is **1 Hz** and the width of the pulse is **20 ns**.
- Increment counter every time one PPS pulses
 - Create a process for a 16-bit **counter** triggered by the 50 MHz **clock** and **enabled** by the pulse above

Simulation vs. Hardware Continued

```
-- increment counter
process(reset, onePPS)
begin
    if(reset = '1') then
        cntr16bit <= (others => '0');
    elsif(onePPS = '1') then
        cntr16bit <= cntr16bit + 1;
    end if;
end process;
```

This is incorrect.

This simulates the same as code below but hardware implementation is different

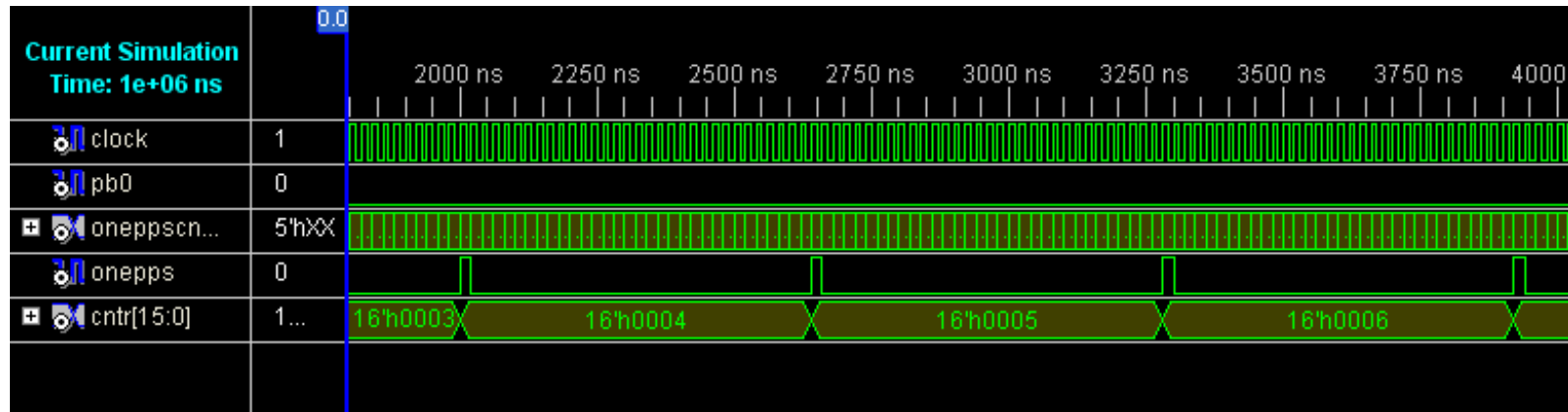
```
-- increment counter
process(reset, clk50)
begin
    if(reset = '1') then
        cntr16bit <= (others => '0');
    elsif(rising_edge(clk50)) then
        if(onePPS = '1') then
            cntr16bit <= cntr16bit + 1;
        end if;
    end if;
end process;
```

VHDL code that generates the onePPS enable signal is omitted to save space

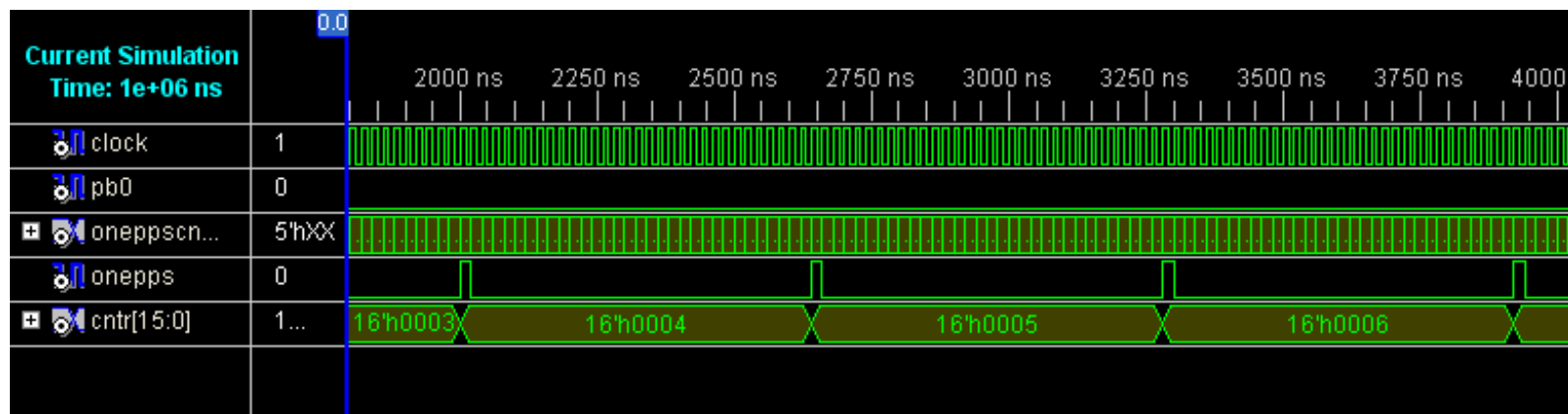
This code is correct.

Simulation will match hardware

Simulation vs. Hardware Continued (2)



Simulation results are the same. Hardware results are different.



Note: 1PPS rate increased for simulation

Simulation vs. Hardware Demo

Demo for one PPS counter
using **good** coding style

Simulation vs. Hardware Demo Continued

Demo for one PPS counter
using **incorrect** coding style